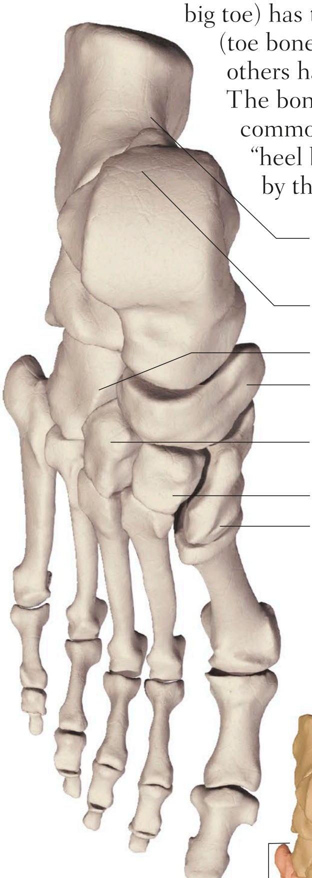
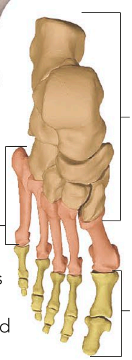
ANKLE AND FOOT

The ankle and foot have a similar bone arrangement to the wrist and hand, except that there are only seven tarsal (ankle) bones. The build of the ankle and foot bones is heavier, for strength and weight-bearing stability. The sole is supported by the five metatarsal bones. The hallux (first digit or big toe) has two phalanges (toe bones), and the others have three each. The bony prominence commonly called the “heel bone” is formed by the calcaneus.

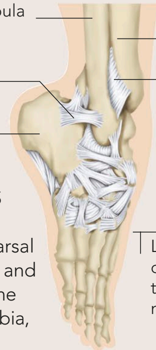




FOOT BONES
The three main bone groups in the foot are the tarsus (ankle); metatarsus (sole); and phalanges (toes or digits).

LIGAMENTS

Ligaments are strong bands or straps of fibrous tissue that provide support to the bones and link bone ends together in and around joints. Ligaments are made of collagen – a tough, elastic protein. A large number of ligaments bind together the complex wrist and ankle joints. Each ligament is named after the bones it links; for example, the calcaneofibular ligament links the calcaneus (“heel bone”) and the fibula.



ANKLE LIGAMENTS
More than a dozen ligaments bind the tarsal bones to each other, and ligaments run from the tarsals to the fibula, tibia, and metatarsals.

WALKING PRESSURE

With each step, the weight of the body moves from the rear to the front of the foot. The heel bears the initial pressure as the foot is put down. The force passes along the arch, which transfers energy and pressure to the ball of the foot, and finally to the big toe for the push-off.

LOAD AREAS ON THE FOOT
These footprint impressions show (from left to right) how the body’s weight transfers from the heel to the ball to the big toe when walking.